

The Repetition Code

From majority vote to Shor's nine-qubit code

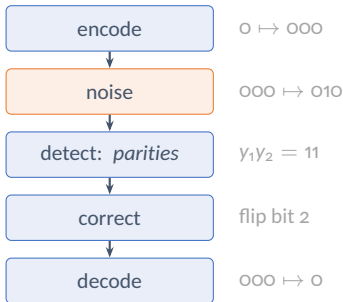
Lecture 12

Term 261

Agenda

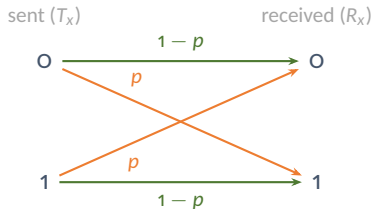
- 1 The classical 3-bit repetition code: majority vote, $p_L = 3p^2 - 2p^3$
- 2 Decoding with parity checks instead of a vote
- 3 The quantum bit-flip code: encoder, syndrome, correction
- 4 The phase-flip code: the same code in the Hadamard basis
- 5 Shor's nine-qubit code: both at once
- 6 Why correcting X, Y, Z is enough

every code today has this shape



A noisy channel flips bits

Binary symmetric (bit-flip) channel



Each bit is flipped independently with probability p : the **bit error rate** (BER).

Example: $p = 10^{-2}$

Send 1000 bits $\Rightarrow$ about **10** arrive wrong.

1000 bits = 125 characters of text: up to 10 corrupted letters.

- The same model describes **memory**: a stored bit flips with probability p
- Lowering p itself is expensive. Can we lower the *effective* error rate instead?

The 3-bit repetition code

- **Logical bit:** the value we want to send
- **Physical bits:** its representation on the channel

Encoding

$0_L \mapsto 000$ $1_L \mapsto 111$

Majority vote: output the value held by at least two of the three bits.

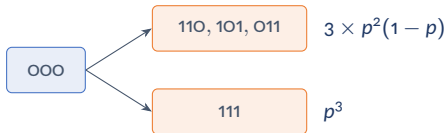
At most **one** flip $\Rightarrow$ the vote is always right.

Takeaway: Redundancy turns “one flip destroys the message” into “it takes two flips”.

sent	received	flips	decoded
000	000	0	0_L
	001, 010, 100	1	0_L
	011, 101, 110	2	1_L (wrong)
	111	3	1_L (wrong)
111	111	0	1_L
	110, 101, 011	1	1_L
	100, 010, 001	2	0_L (wrong)
	000	3	0_L (wrong)

Worked example: how often does the vote fail?

Send $o_L = 000$. The vote fails on two or three flips:



$$p_L = p^3 + 3p^2(1 - p) = 3p^2 - 2p^3$$

Errors now need two independent events:
 $p_L \propto p^2$.

$$p = 10^{-2}$$

$$p_L = 3(10^{-4}) - 2(10^{-6}) = 2.98 \times 10^{-4}$$

1000 bits, no code

10 errors

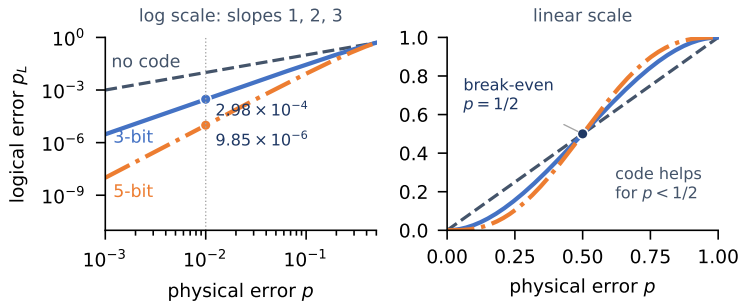
1000 logical bits, coded

0.298 errors

- $33.6\times$ fewer errors ...
- ... for $3\times$ the bits: rate $R = k/n = 1/3$

Takeaway: The code wins whenever $3p^2 - 2p^3 < p$. Next: by how much, and up to which p ?

More repetition, fewer errors — if $p < 1/2$



5-bit code fails on ≥ 3 flips:

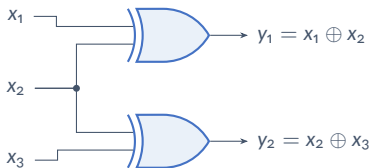
$$p_L = \sum_{j=3}^5 \binom{5}{j} p^j (1-p)^{5-j}$$

At $p = 0.01$: $p_L = 9.85 \times 10^{-6}$, about $10^3 \times$ below p .

- n -bit code: $p_L \sim p^{(n+1)/2}$ (slope on the log plot)
- $p = 1/2$: a coin flip for every n — **break-even**; above it the vote hurts

Decoding without a vote: parity checks

Compute two parities with XOR gates:



received	y_1y_2	flip	result
000 or 111	00	—	unchanged
100 or 011	10	bit 1	000 / 111
010 or 101	11	bit 2	000 / 111
001 or 110	01	bit 3	000 / 111

- A codeword has equal bits, so
 $y_1 = e_1 \oplus e_2$, $y_2 = e_2 \oplus e_3$
- The syndrome y_1y_2 depends only on the **error** e , never on the message

Checked on all 16 (codeword, error) pairs: this decoder outputs exactly what the majority vote outputs, so $p_L = 2.98 \times 10^{-4}$ again.

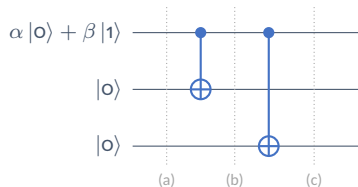
Takeaway: We learned *which bit flipped* without learning *what was sent*. That is the idea that survives into the quantum world.

Going quantum: three obstacles, three answers

Classical step	Quantum obstacle (L10)	Today's answer
copy the bit: $0 \mapsto 000$	no-cloning: no unitary maps $ \psi\rangle \mapsto \psi\rangle^{\otimes 3}$	entangle , do not copy: $\alpha 000\rangle + \beta 111\rangle$
read the bits, vote	measuring collapses $\alpha 0\rangle + \beta 1\rangle$	measure only parities Z_1Z_2, Z_2Z_3
only bit flips occur	phase flips Z , and a continuum of errors (L11)	phase-flip code, Shor's code , and the syndrome discretizes errors

 The parity-check decoder of the last slide never looked at the message. Which of the three obstacles does that already defuse?

Encoding a qubit: entangle, do not copy



$$(a) \alpha |000\rangle + \beta |100\rangle$$

$$(b) \alpha |000\rangle + \beta |110\rangle \quad \text{CNOT } 1 \rightarrow 2$$

$$(c) \alpha |000\rangle + \beta |111\rangle \quad \text{CNOT } 1 \rightarrow 3$$

Logical basis: $|0_L\rangle = |000\rangle$, $|1_L\rangle = |111\rangle$.

Not three copies. Compare

$$|\psi\rangle^{\otimes 3} = \alpha^3 |000\rangle + \alpha^2\beta (|001\rangle + |010\rangle + |100\rangle) \\ + \alpha\beta^2 (|011\rangle + |101\rangle + |110\rangle) + \beta^3 |111\rangle$$

- Cubic in α, β : no unitary produces it for every $|\psi\rangle$ (no-cloning)
- The encoder is linear: two amplitudes in, the same two out
- One qubit of the code alone is $\rho_1 = \text{diag}(|\alpha|^2, |\beta|^2)$: it holds no copy of the relative phase

Takeaway: The information lives in the correlations among the three qubits, not in any one of them.

Bit-flip errors, and how *not* to look for them

error	state	Z_1Z_2	Z_2Z_3	recovery
none	$\alpha 000\rangle + \beta 111\rangle$	+1	+1	—
X_1	$\alpha 100\rangle + \beta 011\rangle$	-1	+1	X_1
X_2	$\alpha 010\rangle + \beta 101\rangle$	-1	-1	X_2
X_3	$\alpha 001\rangle + \beta 110\rangle$	+1	-1	X_3

$Z_1Z_2|x_1x_2x_3\rangle = (-1)^{x_1 \oplus x_2}|x_1x_2x_3\rangle$: the eigenvalue is the classical parity y_1 . Both terms of every row share it, so each row is an **eigenstate** of both checks.

Takeaway: The four errors move the code into four orthogonal subspaces; the two parities say which one.

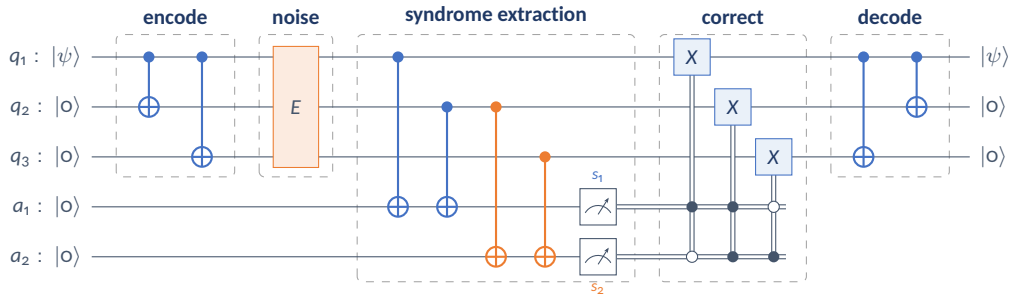
Wrong: measure the qubits

- After X_2 , reading all three gives 010 (prob. $|\alpha|^2$) or 101
- Either way the state collapses: α, β are gone

Right: measure the parities

- Ask only “do qubits 1 and 2 agree?” and “do 2 and 3 agree?”
- The answer is certain, and the state is left untouched

The bit-flip code, end to end



- a_1 collects $x_1 \oplus x_2$, a_2 collects $x_2 \oplus x_3$:
reading s_i measures Z_1Z_2, Z_2Z_3
- Data qubits are **never measured**

- Correction: $s_1s_2 = 10 \rightarrow X_1$, $11 \rightarrow X_2$,
 $01 \rightarrow X_3$
(● = bit is 1, ○ = bit is 0)
- Decode = encoder run backwards

Worked trace: an X on qubit 2

encoded $(\alpha |000\rangle + \beta |111\rangle) |00\rangle_a$
noise X_2 $(\alpha |010\rangle + \beta |101\rangle) |00\rangle_a$
CNOTs into a_1 $\alpha |010\rangle |10\rangle + \beta |101\rangle |10\rangle$
CNOTs into a_2 $\alpha |010\rangle |11\rangle + \beta |101\rangle |11\rangle$
 $= (\alpha |010\rangle + \beta |101\rangle) \otimes |11\rangle_a$
measure a $s_1 s_2 = 11$ with probability 1
apply X_2 $\alpha |000\rangle + \beta |111\rangle$ ✓

Why nothing leaks

Both branches carry the same error, hence the same parities. The ancillas end in a **product state** with the data, so measuring them reveals the error and nothing about α, β .

Simulated (random α, β): X_1, X_2, X_3 give $s = 10, 11, 01$ with probability 1; fidelity after correction 1.

The catch: measuring q_1 instead gives 0 with probability $|\alpha|^2$ and leaves $|010\rangle$: β erased for good.

What the bit-flip code cannot do

1. Two flips: a wrong “correction”

$$X_1X_2 \quad \alpha |110\rangle + \beta |001\rangle$$

syndrome $(+1, -1) \Rightarrow$ “it was X_3 ”

$$\text{apply } X_3 \quad \alpha |111\rangle + \beta |000\rangle$$

A **logical** flip $\bar{X} = X_1X_2X_3$, reported as success.

Probability $3p^2 - 2p^3$, exactly as classically.

2. A phase flip on *any* qubit

$$Z_k(\alpha |000\rangle + \beta |111\rangle) = \alpha |000\rangle - \beta |111\rangle$$

- Z_k commutes with Z_1Z_2 and Z_2Z_3 : syndrome $(+1, +1)$
- Undetected, uncorrected: the logical phase is destroyed
- One Z suffices: there is no majority for a sign

The catch: the bit-flip code corrects one kind of error only; real qubits suffer both (L11).

A phase flip is a bit flip in the Hadamard basis

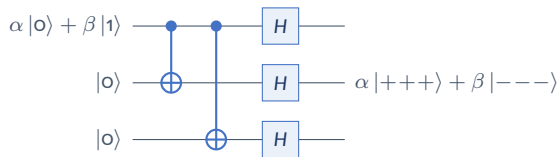
$$Z|\pm\rangle = |\mp\rangle, \quad HZH = X$$

So swap the roles: encode in $\{|+\rangle, |-\rangle\}$,

$$\alpha|0\rangle + \beta|1\rangle \mapsto \alpha|+++ \rangle + \beta|--- \rangle$$

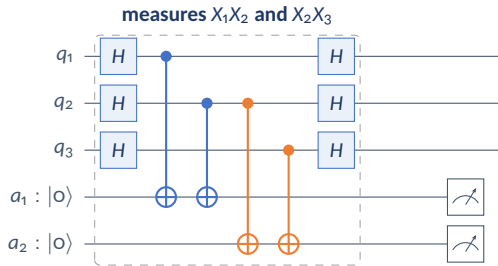
error	state
none	$\alpha +++ \rangle + \beta --- \rangle$
Z_1	$\alpha - ++ \rangle + \beta + -- \rangle$
Z_2	$\alpha + - + \rangle + \beta - + - \rangle$
Z_3	$\alpha ++ - \rangle + \beta -- + \rangle$

Encoder: bit-flip encoder, then H on each qubit



- Logical basis $|0_L\rangle = |+++ \rangle$, $|1_L\rangle = |--- \rangle$
- Same table as before with $0, 1 \rightarrow +, -$ and $X \rightarrow Z$

The phase-flip code: detect in the X basis



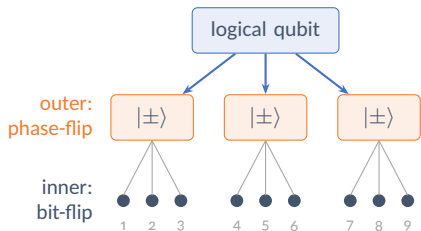
H turns $|\pm\rangle$ into $|0\rangle / |1\rangle$, the old circuit reads the Z -parities, H turns back. Since $HZH = X$: this measures X_1X_2 and X_2X_3 .

X_1X_2	X_2X_3	recovery
+1	+1	—
-1	+1	Z_1
-1	-1	Z_2
+1	-1	Z_3

	bit-flip	phase-flip
checks	Z_1Z_2, Z_2Z_3	X_1X_2, X_2X_3
corrects	one X	one Z
blind to	Z	X

Simulated: Z_1, Z_2, Z_3 read 10, 11, 01; fidelity 1 after recovery.

Shor's code: protect against both by nesting



Step 1 (phase-flip code): $\alpha |+++ \rangle + \beta |-- - \rangle$

Step 2 (bit-flip code on each qubit):

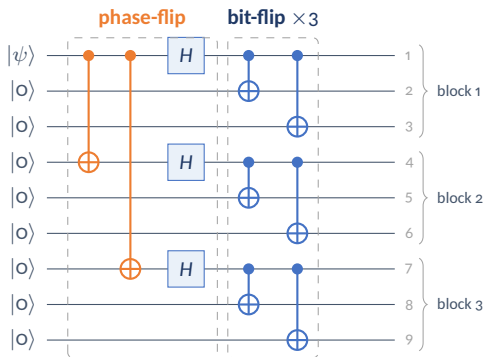
$$|+\rangle \rightarrow \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle), \quad |-\rangle \rightarrow \frac{1}{\sqrt{2}}(|000\rangle - |111\rangle)$$

Shor's code: 1 logical qubit in 9 physical qubits

$$|0_L\rangle = \frac{1}{2\sqrt{2}}(|000\rangle + |111\rangle)^{\otimes 3}, \quad |1_L\rangle = \frac{1}{2\sqrt{2}}(|000\rangle - |111\rangle)^{\otimes 3}$$

$|0_L\rangle$ expanded: 8 terms $|00000000\rangle, \dots, |11111111\rangle$, each with amplitude $\frac{1}{2\sqrt{2}}$.

Shor's encoder

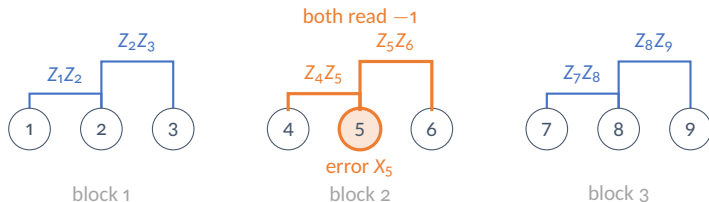


State after each stage

- 1 CNOT $1 \rightarrow 4, 1 \rightarrow 7$:
 $\alpha |000\rangle + \beta |111\rangle$ on 1, 4, 7
- 2 H on 1, 4, 7:
 $\alpha |+++ \rangle + \beta |-- - \rangle$
- 3 CNOTs $1 \rightarrow 2, 3; 4 \rightarrow 5, 6$;
 $7 \rightarrow 8, 9$:
 $\alpha |0_L\rangle + \beta |1_L\rangle$

Checked: the circuit maps $|0\rangle, |1\rangle$ to the codewords of the previous slide exactly (512-dim statevector).

Correcting X errors: three bit-flip codes



- Each block $\frac{1}{\sqrt{2}}(|000\rangle \pm |111\rangle)$ is a bit-flip codeword
- Six checks, two per block, read exactly as before
- An X in block b fires only the checks of block b

Worked: X_5

$Z_4Z_5 = -1$, $Z_5Z_6 = -1$, all other Z -checks $+1$

$\Rightarrow$ middle qubit of block 2 $\Rightarrow$ apply X_5 .

Correcting Z errors: compare the signs of the blocks

A Z on any qubit of a block flips that block's sign:

$$Z_5 : (|000\rangle + |111\rangle)_{456} \rightarrow (|000\rangle - |111\rangle)_{456}$$

$X^{\otimes 3}$ reads a block's sign:

$X^{\otimes 3}(|000\rangle \pm |111\rangle) = \pm(|000\rangle \pm |111\rangle)$. So

$X_1 \cdots X_6$ compares the signs of blocks 1, 2 and

$X_4 \cdots X_9$ those of blocks 2, 3: the phase-flip code's table, one level up.

$X_1 \cdots X_6$	$X_4 \cdots X_9$	sign flipped in
-1	+1	block 1 $\rightarrow$ apply Z_1
-1	-1	block 2 $\rightarrow$ apply Z_4
+1	-1	block 3 $\rightarrow$ apply Z_7

Worked: Z_5

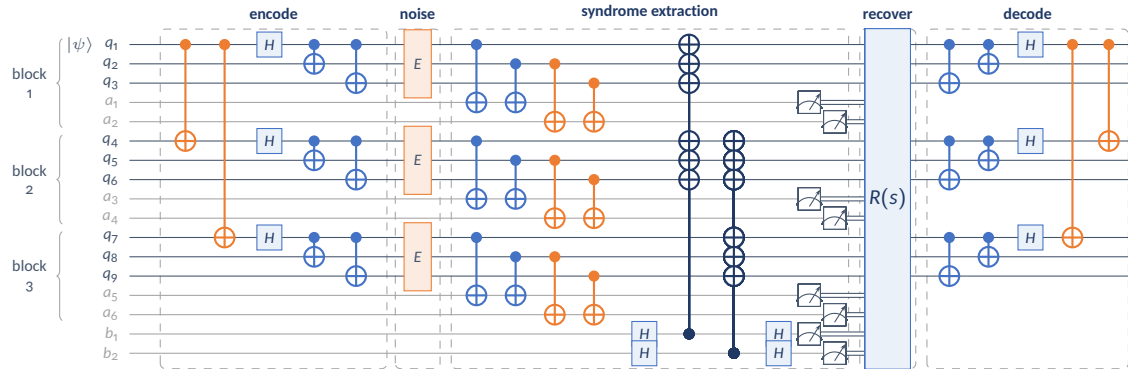
$X_1 \cdots X_6 = -1$, $X_4 \cdots X_9 = -1$, all six Z-checks $+1 \Rightarrow$ block 2 $\Rightarrow$ apply Z_4 . Fidelity 1.

Worked: $Y_5 = iX_5Z_5$

Both kinds fire: Z_4Z_5 , Z_5Z_6 , $X_1 \cdots X_6$, $X_4 \cdots X_9$ all $-1 \Rightarrow$ apply X_5 and Z_4 . Fidelity 1.

Applying Z_4 for a Z_5 error is not a mistake — see two slides on.

Shor's code: the full circuit



Z-checks: under each block, two ancillas read its ZZ parities — the bit-flip circuit, three blocks in parallel. **X-checks:** $b_1 = X_1 \cdots X_6$, $b_2 = X_4 \cdots X_9$ (H, one control fanned out to six targets, H).

$R(s)$: the X and Z tables of the last two slides. 17 qubits; 24 CNOTs per round. Simulated gate for gate: no error and all 27 single-qubit errors return $|\psi\rangle |0 \cdots 0\rangle$.

All 27 single-qubit errors, simulated

qubit	X_k	Z_k	Y_k
1	100000 00	000000 10	100000 10
2	110000 00	000000 10	110000 10
3	010000 00	000000 10	010000 10
4	001000 00	000000 11	001000 11
5	001100 00	000000 11	001100 11
6	000100 00	000000 11	000100 11
7	000010 00	000000 01	000010 01
8	000011 00	000000 01	000011 01
9	000001 00	000000 01	000001 01

bit order $Z_1Z_2, Z_2Z_3, Z_4Z_5, Z_5Z_6, Z_7Z_8, Z_8Z_9 \mid X_{1-6}, X_{4-9}; 1 = \text{eigenvalue}$

—1

The experiment (L12_shor.py)

- Encode random α, β with the circuit
- Apply each of the 27 Paulis
- Compute the 8 parity eigenvalues
- Correct by the two tables

Result: fidelity = 1 for all 27.

- $Y_k = \text{the } X_k \text{ bits} + \text{the } Z_k \text{ bits}$
- Only **21** distinct syndromes:
 Z_1, Z_2, Z_3 share one

Any error, not just X, Y, Z

Any one-qubit operator is a sum of Paulis,
 $E = e_0 \mathbb{I} + e_1 X + e_2 Y + e_3 Z$, so an error on
qubit k gives a **superposition**

$$E_k |\psi_L\rangle = e_0 |\psi_L\rangle + e_1 X_k |\psi_L\rangle \\ + e_2 Y_k |\psi_L\rangle + e_3 Z_k |\psi_L\rangle .$$

The four terms have different syndromes.
Measuring the syndrome **collapses** onto one
term, with probability $|e_j|^2$ — then we correct
it.

Demo: rotation $e^{-i\theta \mathbf{n} \cdot \boldsymbol{\sigma}}$ on qubit 3

$\theta = 0.3$, $\mathbf{n} = (1, 2, 2)/3$:

$e_0 = \cos \theta$, $(e_1, e_2, e_3) = -i \sin \theta \mathbf{n}$

syndrome	prob.	correction	fidelity
000000 00	0.9127	none	1
010000 00	0.0097	X_3	1
010000 10	0.0388	X_3, Z_1	1
000000 10	0.0388	Z_1	1

Probabilities = $\cos^2 \theta$ and $\sin^2 \theta \cdot (\frac{1}{9}, \frac{4}{9}, \frac{4}{9})$. Without
correction the fidelity is 0.913.

Takeaway: Correct X, Y, Z on every qubit and you have corrected every single-qubit error.

A quantum surprise: different errors, same effect

Z_1, Z_2, Z_3 give the same syndrome — so how can the decoder correct them all?

Because they do the **same thing** to every codeword. $Z_1 Z_2$ is a parity check, so $Z_1 Z_2 |\psi_L\rangle = |\psi_L\rangle$, and

$$Z_2 |\psi_L\rangle = Z_1 (Z_1 Z_2) |\psi_L\rangle = Z_1 |\psi_L\rangle.$$

Applying Z_1 after a Z_2 error leaves $Z_1 Z_2$, which acts as the identity on the code.

Takeaway: Degeneracy has no classical analogue. L13 makes it precise (Knill–Laflamme); L14 explains it with the stabilizer group.

Checked numerically

- $Z_1 |\psi_L\rangle = Z_2 |\psi_L\rangle = Z_3 |\psi_L\rangle$
- $Z_4 |\psi_L\rangle = Z_5 |\psi_L\rangle, \dots$
- but $Z_1 |\psi_L\rangle \neq Z_4 |\psi_L\rangle$

Classically, two different errors must be told apart. Here they need not be: the code is **degenerate**.

What Shor's code costs, and where it breaks

	bit-flip	phase-flip	Shor
qubits n	3	3	9
parity checks	2	2	8
CNOTs/round	4	4	24
corrects	one X	one Z	any 1-qubit error
rate $R = k/n$	$1/3$	$1/3$	$1/9$

Two errors: 180 of the 324 two-qubit Paulis are still corrected — the other 144 are not.

■ *Nine qubits to protect one. L13 asks: can we do better? (Yes: 7 qubits, then 5.)*

Two errors in the wrong places

- X_1X_2 (same block): decoder applies X_3 ; net $X_1X_2X_3$ acts as a logical $\bar{Z}$
- Z_1Z_4 (different blocks): decoder blames block 3, applies Z_7 ; net $Z_1Z_4Z_7$ acts as a logical $\bar{X}$

Both fail *silently*: the syndrome looked like a single error.

Summary

- Repetition with majority vote: $p_L = 3p^2 - 2p^3$; at $p = 0.01$, 2.98×10^{-4} . Five bits: 9.85×10^{-6} . Useful only for $p < 1/2$
- Parity checks find *which bit* flipped without reading the message — the classical idea that survives
- Bit-flip code: $\alpha |000\rangle + \beta |111\rangle$ (entangled, not copied); measure Z_1Z_2, Z_2Z_3 with ancillas; correct one X
- Phase-flip code: the same in the Hadamard basis; measure X_1X_2, X_2X_3 ; correct one Z
- Shor's code nests the two: 9 qubits, 8 checks, corrects any single-qubit error
- The syndrome measurement collapses any error onto a Pauli: correcting X, Y, Z suffices
- Different errors can act identically on the code (degeneracy) — L13, L14

References

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- M. A. Nielsen, I. L. Chuang (2010). *Quantum Computation and Quantum Information*, 10th anniversary ed. Cambridge University Press — §10.1 (the three-qubit codes, Shor’s code), §10.2 (discretization of errors).
- S. J. Devitt, W. J. Munro, K. Nemoto (2013). “Quantum error correction for beginners.” *Rep. Prog. Phys.* **76**, 076001. [arXiv:0905.2794](#).
- D. Gottesman (2009). “An introduction to quantum error correction and fault-tolerant quantum computation.” [arXiv:0904.2557](#).

All numbers and syndrome tables on these slides were computed for this course by statevector simulation (scripts `L12_repetition.py`, `L12_three_qubit.py`, `L12_shor.py`).